📁 1. Project Structure (Maven)

```
library-system/
├── pom.xml
├── Jenkinsfile
└── src/
    ├── main/
    │   └── java/com/library/App.java
    └── test/
        └── java/com/library/AppTest.java
```

📦 2. pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
         xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
         xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
         http://maven.apache.org/xsd/maven-4.0.0.xsd">

  <modelVersion>4.0.0</modelVersion>

  <groupId>com.library</groupId>
  <artifactId>library-system</artifactId>
  <version>1.0-SNAPSHOT</version>

  <properties>
    <maven.compiler.source>17</maven.compiler.source>
    <maven.compiler.target>17</maven.compiler.target>
  </properties>

  <dependencies>
    <dependency>
      <groupId>org.junit.jupiter</groupId>
      <artifactId>junit-jupiter</artifactId>
      <version>5.10.0</version>
      <scope>test</scope>
    </dependency>
  </dependencies>

  <build>
    <plugins>
```

```
        <plugin>
          <groupId>org.apache.maven.plugins</groupId>
          <artifactId>maven-surefire-plugin</artifactId>
          <version>3.2.5</version>
        </plugin>
      </plugins>
    </build>

</project>
```

🧠 **3. App.java**

```
package com.library;

import java.util.*;

class User {
  String name;
  int id;
  int books;
  int days;

  static final int ALLOWED_DAYS = 7;
  static final int FINE_PER_DAY = 10;

  User(String name, int id, int books, int days) {
    this.name = name;
    this.id = id;
    this.books = books;
    this.days = days;
  }

  int calculateFine() {
    if (days > ALLOWED_DAYS) {
      return (days - ALLOWED_DAYS) * FINE_PER_DAY;
    }
    return 0;
  }
}
```

```
public class App {
   public static void main(String[] args) {

      Scanner sc = new Scanner(System.in);

      System.out.println("Enter Name:");
      String name = sc.next();

      System.out.println("Enter ID:");
      int id = sc.nextInt();

      System.out.println("Enter number of books:");
      int books = sc.nextInt();

      System.out.println("Enter number of days:");
      int days = sc.nextInt();

      User user = new User(name, id, books, days);

      System.out.println("\n--- User Details ---");
      System.out.println("Name: " + user.name);
      System.out.println("ID: " + user.id);
      System.out.println("Books Issued: " + user.books);
      System.out.println("Days: " + user.days);

      int fine = user.calculateFine();

      if (fine > 0) {
         System.out.println("Late Fine: ₹" + fine);
      } else {
         System.out.println("No Fine");
      }
   }
}
```

🧪 4. AppTest.java

```
package com.library;

import org.junit.jupiter.api.Test;
```

```
import static org.junit.jupiter.api.Assertions.*;

class AppTest {

  @Test
  void testNoFine() {
    User u = new User("Test", 1, 2, 5);
    assertEquals(0, u.calculateFine());
  }

  @Test
  void testWithFine() {
    User u = new User("Test", 1, 2, 10);
    assertEquals(30, u.calculateFine());
  }
}
```

🔧 5. Git Commands

```
git init
git add .
git commit -m "Initial commit - Library System"
```

Create repo on GitHub

```
git remote add origin https://github.com/your-username/library-system.git
git branch -M main
git push -u origin main
```

💻 6. Clone + Modify

```
git clone https://github.com/your-username/library-system.git
cd library-system
code .
```

Make change → then:

```
git add .
git commit -m "Updated logic"
git push
```

🧪 7. Build & Test

mvn clean install

👉 Must show:

BUILD SUCCESS

🚀 8. Jenkinsfile

```
pipeline {
  agent any

  stages {

    stage('Clone') {
      steps {
        git 'https://github.com/your-username/library-system.git'
      }
    }

    stage('Build') {
      steps {
        bat 'mvn clean compile'
      }
    }

    stage('Test') {
      steps {
        bat 'mvn test'
      }
    }

    stage('Package') {
      steps {
        bat 'mvn package'
      }
    }
  }
}
```

👉 Linux → replace bat with sh

⚙️ 9. Jenkins Setup

In Jenkins:

- New Item → Pipeline
- Paste Jenkinsfile
- Click **Build Now**